

Hadoop HA Cluster Documentation

(Docker-Based)

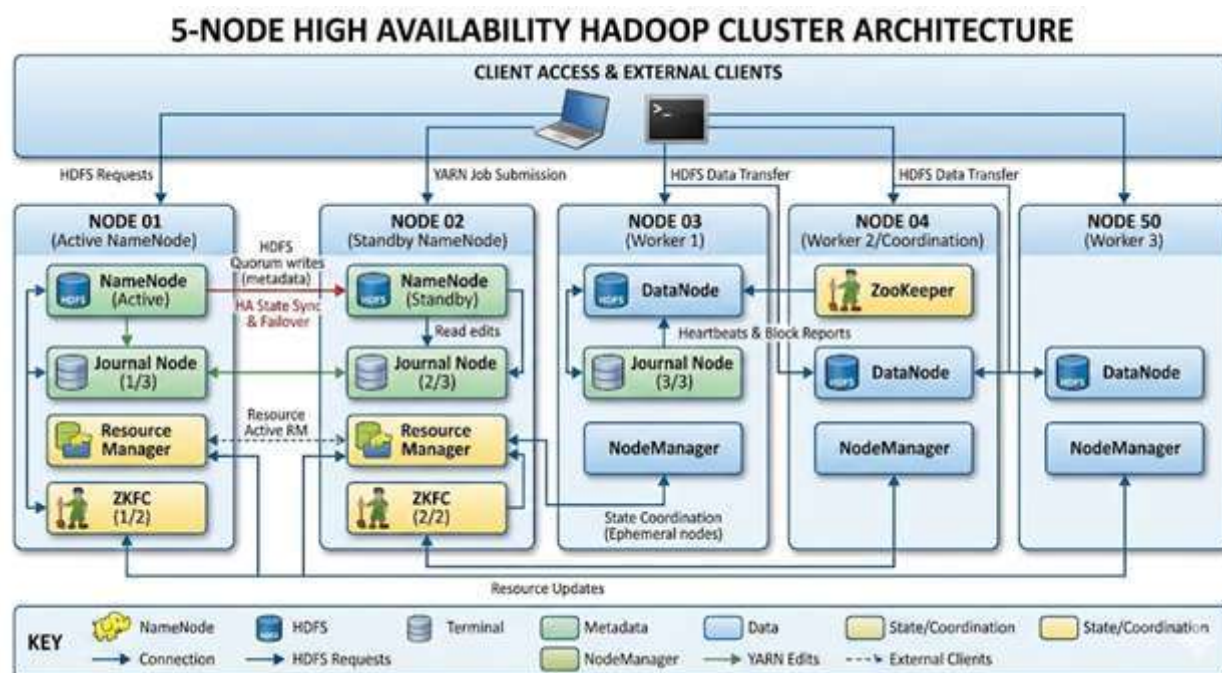
Github Link: https://github.com/FaridaEmad/Hadoop_Cluster.git

1. Cluster Architecture Overview

This project builds a **High Availability (HA) Hadoop Cluster** using Docker containers.

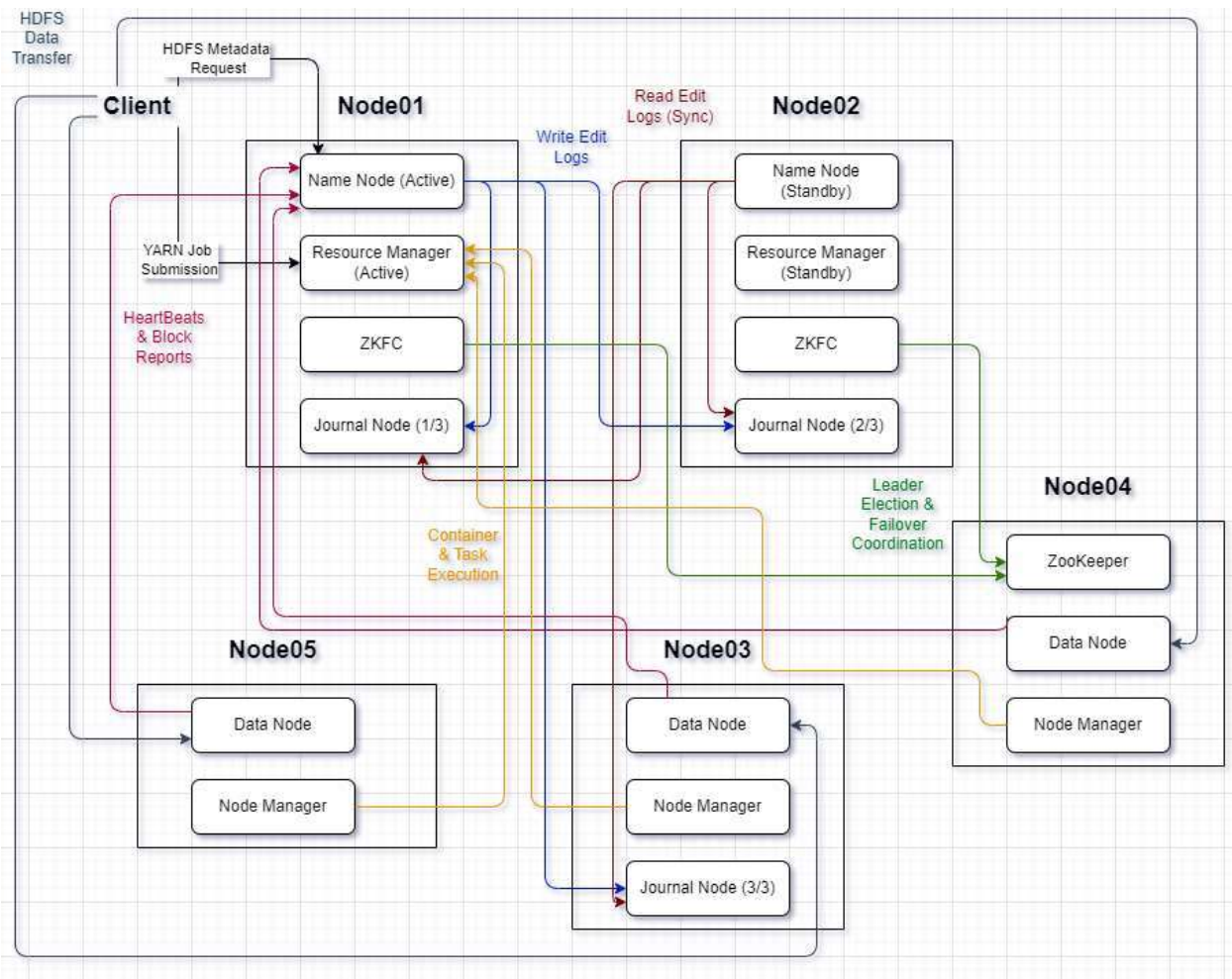
- HDFS High Availability (2 NameNodes)
- JournalNodes (3 nodes)
- ZooKeeper-based automatic failover
- YARN High Availability (2 ResourceManagers)
- 3 DataNodes
- 1 ZooKeeper node

2. Nodes & Roles Distribution



Note: this picture is generated using ai to visualize my design.

Note: I drew the following design.



Node	Roles	Why this Design?
Node01	NameNode (Active), JournalNode, ZKFC, ResourceManager	Primary master node
Node02	NameNode (Standby), JournalNode, ZKFC, ResourceManager	HA standby master
Node03	DataNode, JournalNode, NodeManager	Worker + Journal
Node04	ZooKeeper, DataNode, NodeManager	Worker + Zookeeper
Node05	DataNode, NodeManager	Worker

3. HDFS High Availability Design

NameNodes

- **node01** --> **nn1**
- **node02** --> **nn2**

Why?

- Prevents single point of failure
- If Active NN fails -> Standby becomes Active automatically

JournalNodes (Quorum: 3 nodes)

Running on:

- node01
- node02
- node03

Why 3?

- Quorum-based system requires **odd number**
- Majority (2/3) needed for consistency
- Prevents split-brain problem

Shared edits:

qjournal://node01:8485;node02:8485;node03:8485/hadoopproject

ZooKeeper & ZKFC

- ZooKeeper runs on **node04**
- ZKFC runs on node01 & node02

ZooKeeper responsibilities:

- Leader election
- Failover coordination
- Lock management

4. Data Layer (Storage)

DataNodes:

- node03
- node04
- node05

Why 3 DataNodes?

- Distributed storage
- Block replication (but in the project description we was asked to set the replication factor = 1)
- Fault tolerance
- Parallel processing

5. YARN High Availability

I have implemented:

- ResourceManager HA
- 2 RMs:
 - rm1 -> node01
 - rm2 -> node02
- ZooKeeper-based leader election

NodeManagers run on:

- node03
- node04
- node05

Why this design?

- Prevents YARN master failure
- Applications continue running even if one RM fails

6. Docker Architecture Design

Network

Custom bridge network:

172.30.0.0/16

Static IP assignment:

- Ensures predictable communication
- Required for HA configuration
- Avoids DNS resolution issues

7. Startup Logic Design

You used a **role-based startup script**:

```
case $HOST in
```

Advantages:

- Same Docker image for all nodes
- Role determined by hostname
- Clean, reusable design
- Automated formatting logic
- Automatic standby bootstrap

8. Master vs Worker Separation

Master Layer

- node01
- node02

Handles:

- Metadata
- Scheduling
- Cluster coordination

Worker Layer

- node03
- node04
- node05

Handles:

- Block storage
- Task execution

- Shuffle processing

9. Fault Tolerance Mechanisms

FAILURE	WHAT HAPPENS?
ACTIVE NN CRASHES	Standby promoted automatically
RESOURCEMANAGER FAILS	Other RM becomes active
DATANODE FAILS	Blocks re-replicated
JOURNALNODE FAILS	System continues